

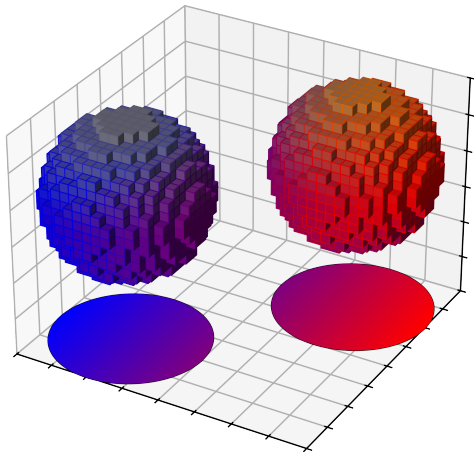
Subspace clustering in high-dimensions

Phase transitions & Statistical-to-Computational gap

Roger Michel Bader Angelo Giustiniani Cheyoung Lee
Stefano Magrini Alunno Ilaria Pellegrinelli
Kazuma Sawaya Sophie Uluatam

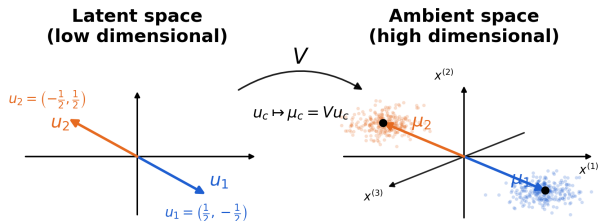
ICTP-INdAM-SLMATH Summer Graduate School for Machine Learning
June 2026

Motivation



- ▶ **Data explosion:** Clustering is essential in modern ML pipelines.
- ▶ **Intrinsic dimensionality:** High-dimensional data often lives in low-dimensional spaces (e.g., MNIST: 784D space $\rightarrow$ ~ 10 D intrinsic).
- ▶ **The Core Question:** When is it *statistically* and *algorithmically* possible to reconstruct these low-dimensional features?

Setup



Low-Rank Mixture Model

$$X = \sqrt{\frac{\lambda}{s}} VU^\top + W$$

- ▶ $X \in \mathbb{R}^{d \times n}$: Data matrix
- ▶ $\lambda > 0$: Signal-to-Noise Ratio (SNR)
- ▶ $s \in \mathbb{N}$: Number of active features (columns)
- ▶ $\rho = s/d$: Density

Model Components

- ▶ **Cluster Codes** ($U \in \mathbb{R}^{n \times k}$) represent low-dimensional, zero-mean cluster assignment vectors.
- ▶ **Sparse Features** ($V \in \mathbb{R}^{d \times k}$) maps assignment vectors to sparse means via $\mu_c = Vu_c$.
- ▶ **Noise** ($W \in \mathbb{R}^{d \times n}$): i.i.d. standard Gaussian matrix.

$$u_c = \frac{1}{k}(-1, \dots, \underbrace{k-1}_{c\text{-th}}, \dots, -1)$$

$$v_i \sim \rho \mathcal{N}(\mathbf{0}, \mathbf{I}_k) + (1 - \rho) \delta_0$$

Goal of the paper

Clustering = Estimating U

The **Bayes-optimal estimator** has a closed-form solution

$$\hat{U}_{\text{bo}} = \arg \min_{U \in \mathbb{R}^{n \times k}} \text{MSE}(U) = \mathbb{E}[U|X], \quad \text{MSE} = \min_{\pi} \frac{1}{n} \mathbb{E} \|\pi(\hat{U}) - U_{\star}\|_F^2,$$

where the posterior distribution is given by

$$P(U | X) = \frac{1}{Z(X)} \prod_{\nu=1}^n P_u(\mathbf{u}_{\nu}) \int_{(\mathbb{R}^k)^d} \prod_{i=1}^d P_v(d\mathbf{v}_i) \prod_{\nu=1}^n \prod_{i=1}^d e^{-\frac{1}{2} \left(X_{\nu i} - \sqrt{\frac{\lambda}{s}} \mathbf{u}_{\nu}^{\top} \mathbf{v}_i \right)^2}$$

Questions

- 1 When is $\hat{U}_{\text{bo}}$ better than a random baseline? $\longrightarrow$ **information-theoretic limit**
- 2 When can a polynomial-time algorithm reach $\hat{U}_{\text{bo}}$? $\longrightarrow$ **algorithmic limit**

Main Result 1

In the limit as $n, d, s \rightarrow \infty$ with fixed ratios $\rho = s/d$ and $\alpha = n/d$, the best possible MSE for recovering cluster assignments reduces to the $k \times k$ optimization problem:

$$\lim_{n \rightarrow \infty} \text{MMSE} = \frac{k-1}{k} - \text{Tr}(M_u^*), \quad M_u^* = \arg \min_{M_u \in \mathbb{R}^{k \times k}} \left\{ \max_{M_v \in \mathbb{R}^{k \times k}} \Phi_{\text{rs}}(M_u, M_v) \right\}. \quad (8)$$

Main result 2

In the limit as $n, d, s \rightarrow \infty$ with fixed ratios $\rho = s/d$ and $\alpha = n/d$, the correlations between ground truth and AMP estimators

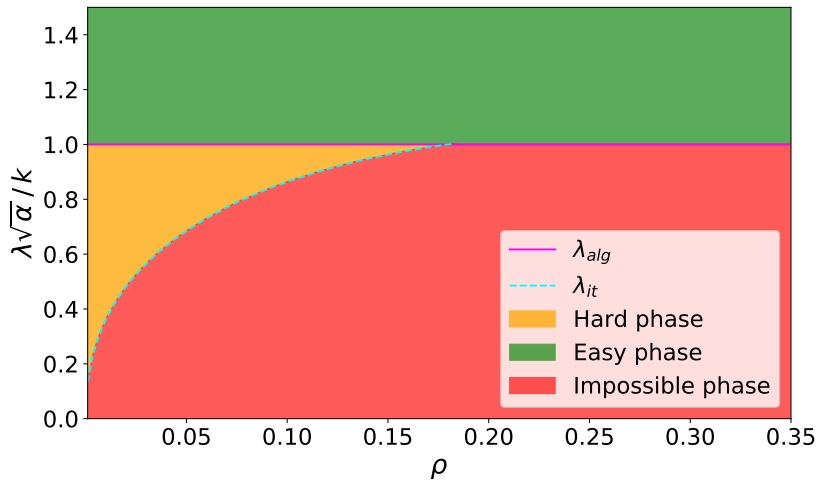
$$M_u^t = \frac{1}{n} \mathbf{U}_\star^\top \hat{\mathbf{U}}_{\text{amp}}^t, \quad M_v^t = \frac{1}{n} \mathbf{V}_\star^\top \hat{\mathbf{V}}_{\text{amp}}^t \quad (15)$$

satisfy the *State-Evolution Equations*

$$M_u^{t+1} = \mathbb{E}_{\mathbf{u}_\star \sim P_u, \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}_k, \mathbf{I}_k)} \left[\eta_u \left(\frac{\alpha \lambda M_u}{\rho}, \frac{\alpha \lambda M_u}{\rho} \mathbf{u}_\star + \sqrt{\frac{\alpha \lambda M_u}{\rho}} \boldsymbol{\xi} \right) \mathbf{u}_\star^\top \right] \quad (16)$$

$$M_v^{t+1} = \mathbb{E}_{\mathbf{v}_\star \sim P_v, \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}_k, \mathbf{I}_k)} \left[\eta_v \left(\frac{\lambda M_v}{\rho}, \frac{\lambda M_v}{\rho} \mathbf{v}_\star + \sqrt{\frac{\lambda M_v}{\rho}} \boldsymbol{\xi} \right) \mathbf{v}_\star^\top \right] \quad (17)$$

Phase Diagram



- ▶ **Theoretical Framework:**

Sparse k-GMM clustering $\longrightarrow$ *Low-rank Matrix Factorization*.

- ▶ **Performance Limits:**

Derived the exact asymptotic MMSE characterization in high dimensions.

- ▶ **The Efficiency Gap:** Identified a sharp *statistical-to-computational* gap:

$$\text{IT Limit} \quad \lambda_{it} \approx \sqrt{\frac{-k\rho \log \rho}{\alpha}}$$

$$\text{Algorithmic Limit} \quad \lambda_{alg} = \frac{k}{\sqrt{\alpha}}$$

- ▶ **Computational Hardness:**

A hard phase emerges when the cluster means are very sparse ($\rho = \frac{s}{d} \ll 1$).

Thanks for the attention!



SCAN ME

Questions?